

API_REFERENCE

HelixTrack API Reference — 2026-06-21

Overview

HelixTrack Core Backend API reference documentation.

Base URL: `http://localhost:8080` **Authentication:** JWT Bearer Token

Authentication

Login

POST /api/v1/auth/login
Content-Type: application/json

```
{
  "username": "string",
  "password": "string"
}
```

Response:

```
{
  "token": "jwt_token",
  "user": {
    "id": "string",
    "username": "string",
    "email": "string"
  }
}
```

Register

POST /api/v1/auth/register
Content-Type: application/json

```
{
  "username": "string",
  "email": "string",
  "password": "string",
  "fullName": "string"
}
```

Projects

List Projects

GET /api/v1/projects
Authorization: Bearer <token>

Get Project

GET /api/v1/projects/:id
Authorization: Bearer <token>

Create Project

POST /api/v1/projects
Authorization: Bearer <token>
Content-Type: application/json

```
{
  "name": "string",
  "key": "string",
  "description": "string",
  "lead": "string"
}
```

Update Project

PUT /api/v1/projects/:id
Authorization: Bearer <token>
Content-Type: application/json

```
{
  "name": "string",
  "description": "string"
}
```

Delete Project

DELETE /api/v1/projects/:id
Authorization: Bearer <token>

Tickets

List Tickets

```
GET /api/v1/tickets?  
project_id=<id>&status=<status>&priority=<priority>  
Authorization: Bearer <token>
```

Get Ticket

```
GET /api/v1/tickets/:id  
Authorization: Bearer <token>
```

Create Ticket

```
POST /api/v1/tickets  
Authorization: Bearer <token>  
Content-Type: application/json
```

```
{  
  "title": "string",  
  "description": "string",  
  "ticketTypeId": "string",  
  "ticketStatusId": "string",  
  "projectId": "string",  
  "priority": "string",  
  "assignee": "string"  
}
```

Update Ticket

```
PUT /api/v1/tickets/:id  
Authorization: Bearer <token>  
Content-Type: application/json
```

```
{  
  "title": "string",  
  "description": "string",  
  "status": "string",  
  "assignee": "string"  
}
```

Delete Ticket

```
DELETE /api/v1/tickets/:id  
Authorization: Bearer <token>
```

Comments

List Comments

GET /api/v1/comments?ticket_id=<id>
Authorization: Bearer <token>

Create Comment

POST /api/v1/comments
Authorization: Bearer <token>
Content-Type: application/json

```
{  
  "ticketId": "string",  
  "content": "string"  
}
```

Users

List Users

GET /api/v1/users
Authorization: Bearer <token>

Get User

GET /api/v1/users/:id
Authorization: Bearer <token>

Teams

List Teams

GET /api/v1/teams?project_id=<id>
Authorization: Bearer <token>

Create Team

POST /api/v1/teams
Authorization: Bearer <token>
Content-Type: application/json

```
{
```

```
"name": "string",  
"description": "string",  
"projectId": "string"  
}
```

WebSocket

Connect

`ws://localhost:8080/ws?token=<jwt_token>`

Events

- `ticket.created` — New ticket created
 - `ticket.updated` — Ticket updated
 - `comment.created` — New comment
 - `user.online` — User came online
 - `user.offline` — User went offline
-

Error Codes

Code	Description
200	Success
201	Created
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
500	Internal Server Error

Cross-references

- [Architecture](#)
- [Implementation Plan](#)

Sources verified 2026-06-22: /Volumes/T7/Projects/helix_track/core/ (sibling HelixTrack source tree) , <https://github.com/Helix-Track/Everything>

REPO-STATE-DERIVED (per §11.4.99 the sources are the cross-referenced HelixTrack source tree, following the docs/ARCHITECTURE.md precedent — HelixTrack is a sibling own-org project, not a third-party live service this doc instructs against). Cross-referenced on 2026-06-22: - The sibling source tree / Volumes/T7/Projects/helix_track/core/ is present locally and is the authoritative source for these REST endpoints / request- response shapes / status codes; auth is JWT-Bearer (confirmed in helix_track/auth/{ARCHITECTURE,README}.md). - The upstream umbrella repo github.com/Helix-Track/Everything confirms a Core backend service exists in the HelixTrack project. - **Negative finding:** the GitHub umbrella README does NOT itemise the individual endpoints — endpoint-level detail is derived from the local core/ tree, which is the source to re-verify against when the backend changes. The localhost:8080 base URL is a dev-default, not a deployed endpoint.